

FleetPath — Fuel Pathway Decision Report

Independent cost and carbon analysis for fleet procurement decisions

Fleet Profile

REGION	VEHICLE CLASS	FLEET SIZE	ANNUAL MILEAGE	LIFECYCLE HORIZON
QC	school_bus_typeC	10 vehicles	12,000 mi	12 yr

TL;DR Verdict

Battery-Electric wins: no capital payback within fleet lifecycle versus the diesel baseline, cutting lifecycle CO2e emissions by 99.8% relative to diesel. Note: Battery-Electric's lifetime cost is \$65,115 CAD higher than diesel's over the fleet lifecycle -- this pathway is favored on blended cost/carbon utility, not on lowest raw cost.

Payback vs. diesel: N/A | Emissions reduction vs. diesel: 99.8%

Pathway Comparison

Fuel Pathway	Lifetime TCO	Lifecycle CO2e	Cold-Climate Adj.
Diesel	\$387,088 CAD	203.2 t	No
Battery-Electric (Winner)	\$452,203 CAD	0.3 t	No
Hydrogen Fuel Cell	\$1,469,067 CAD	0.7 t	No
CNG	\$359,145 CAD	135.6 t	No
Biodiesel (B20)	\$390,293 CAD	166.1 t	No

Methodology: figures derived from NREL H2A hydrogen station cost models, Argonne National Laboratory AFLEET 2023 vehicle and infrastructure baselines, EPA eGRID 2023 Rev 2 regional grid emission factors (US), and Environment and Climate Change Canada National Inventory Report 2023 (NIR 1990-2021) provincial grid intensities (Canada). This is a decision-support estimate, not a procurement guarantee.